

Campus Marketplace

Requirements Specification and Initial Development Schedule

CMSI 4071 Senior Capstone Project

Project	Campus Marketplace
Deliverable	D02 Requirements Specification and Initial Development Schedule
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Document Purpose

This document supplies the Week 5 deliverables for Campus Marketplace. Section 4.4 provides the initial development schedule, including a Gantt chart and task-resource assignments. Section 5.0 specifies the system's functional, performance, development-environment, and execution-environment requirements. The numbered shall statements define testable obligations that will later support design decisions and requirements-based testing.

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4.4 Project Schedule

Campus Marketplace will be developed iteratively across the sixteen-week course schedule. Documentation, design, implementation, testing, integration, deployment, meetings, and status reporting are scheduled as project work rather than treated as incidental activities. The initial schedule protects the final three weeks for integration, correction, deployment, and demonstration preparation.

The schedule assumes the minimum viable product includes verified .edu accounts, campus association, customer and vendor profiles, marketplace listings, search and filters, availability, appointment and product requests, reviews, notifications, and basic administration. Integrated payments, delivery management, native mobile applications, and real-time chat are outside the initial release.

4.4.1 Gantt Chart

Milestone	Course Week	Completion Evidence
D02 Requirements and Initial Schedule	5	Approved SRS draft and baseline schedule
D03 Complete Software Development Plan	7	Resources, organization, and schedule finalized
D04 Architectural Design	10	Interfaces, deployment structure, and data model defined
D05 Detailed Design	12	Component-level behavior and data design completed
D06 Alpha or Beta Review	12 to 13	Integrated critical path demonstrated
D07 User Manual Interim Update	13	README-based usage instructions available
D08 Final Presentation	15	Release candidate demonstrated
D09 Final Product Delivery	16	Code freeze and final documentation committed

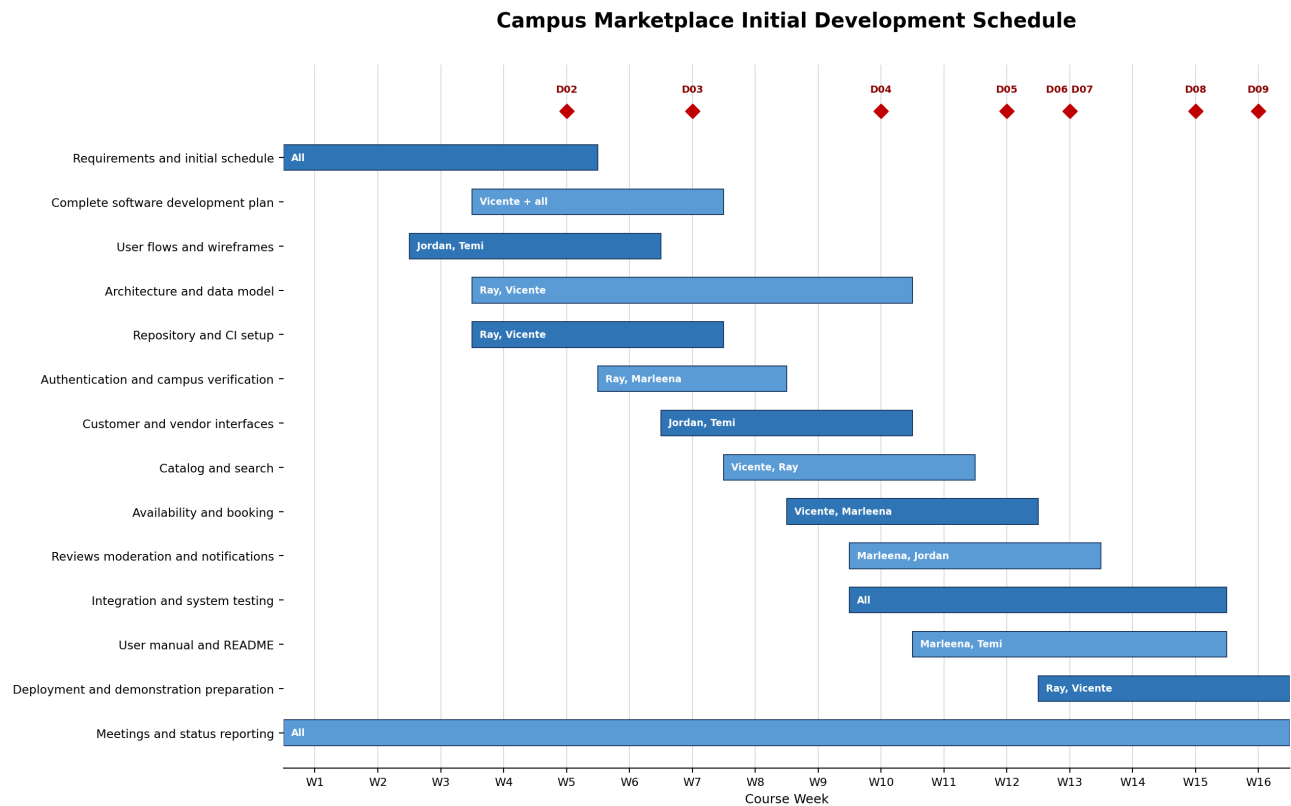


Figure 1 Campus Marketplace initial development schedule

4.4.2 Task Resource Table

The assignments below are the initial resource-loading plan. Owners coordinate the task, while every team member participates in reviews, integration, testing, documentation, and presentations. Named assignments may be rebalanced in the complete Software Development Plan if progress data shows an uneven workload.

ID	Task	Initial Owners	Weeks	Hardware	Software	Exit Criterion
T01	Requirements and initial schedule	Vicente and all reviewers	1 to 5	Team laptops	Word, GitHub	D02 committed
T02	User flows and wireframes	Jordan and Temi	3 to 6	Team laptops	Figma or Canva	Critical flows approved
T03	Architecture and data model	Ray and Vicente	4 to 10	Team laptops	Diagram tool, PostgreSQL	D04 approved
T04	Repository and CI setup	Ray and Vicente	4 to 7	Team laptops	GitHub Actions	Build and tests run on push
T05	Authentication and campus mapping	Ray and Marleena	6 to 8	Laptop and test inbox	FastAPI, Cognito	Verified .edu flow passes
T06	Customer and vendor interfaces	Jordan and Temi	7 to 10	Laptop and test phone	React, TypeScript	Critical screens integrated
T07	Catalog and search	Vicente and Ray	8 to 11	Team laptops	FastAPI, PostgreSQL	Search acceptance tests pass
T08	Availability and booking	Vicente and Marleena	9 to 12	Team laptops	FastAPI, PostgreSQL	Booking critical path passes

ID	Task	Initial Owners	Weeks	Hardware	Software	Exit Criterion
T09	Reviews moderation and notifications	Marleena and Jordan	10 to 13	Laptop and test inbox	React, FastAPI, email service	Trust workflows pass
T10	Integration and system testing	All	10 to 15	Laptops and test phone	Pytest, Vitest, Postman	Release test suite passes
T11	Deployment and monitoring	Ray and Vicente	13 to 16	Team laptops	AWS, GitHub Actions	Release candidate is reachable
T12	User manual and presentation	Marleena, Temi, and all presenters	11 to 16	Laptop and projector	README, presentation software	D08 and D09 complete
T13	Meetings and status reports	All	1 to 16	Team laptops	Calendar, GitHub Issues	Records posted on schedule

5.0 Requirements Specification

5.1 Introduction

Campus Marketplace is a responsive web application that connects verified college students with student-operated businesses associated with their campus. Customers discover vendors, review listings, submit service appointments or product reservations, and provide post-transaction reviews. Vendors publish profiles, listings, prices, images, discounts, availability, and request decisions. Administrators address reported content and account misuse. The system centralizes activity that is currently spread across social media, flyers, group chats, and word of mouth.

The first release excludes integrated payment processing, delivery coordination, native mobile applications, and real-time chat. Customers and vendors will arrange payment and physical fulfillment outside the platform. These boundaries preserve a one-semester scope while retaining a complete discovery and request-management workflow.

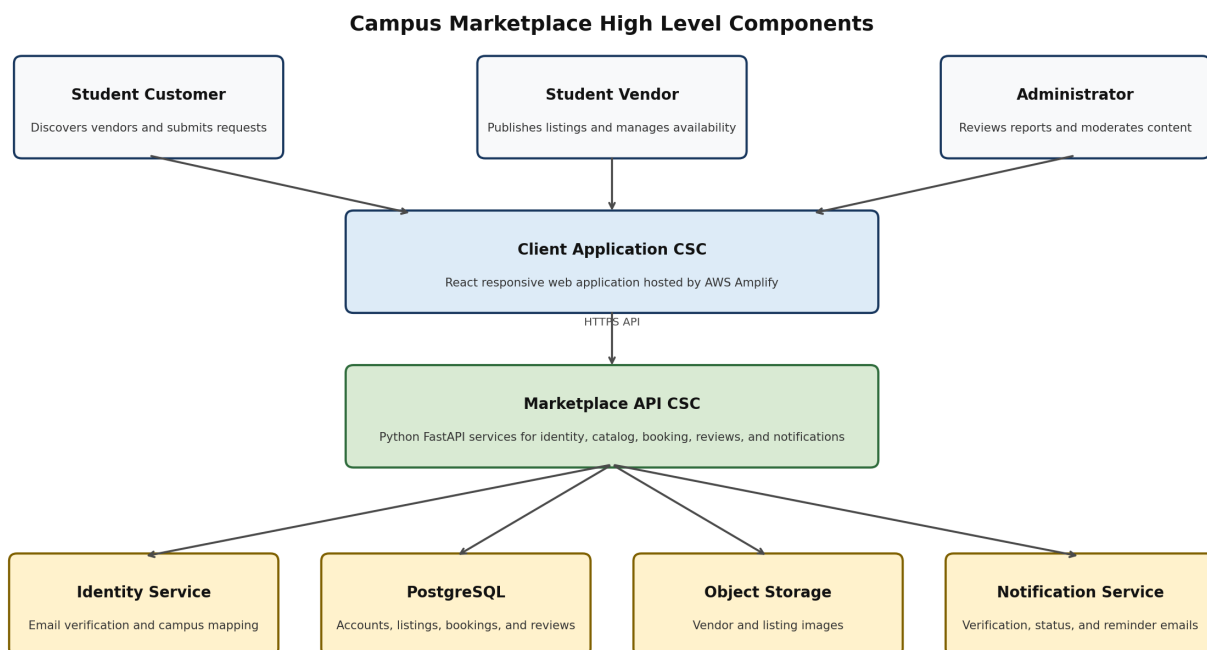


Figure 2 Campus Marketplace high level system components

Section 5.2 identifies the Campus Marketplace CSCI and its component hierarchy. Section 5.3 defines functional requirements grouped by CSC. Section 5.4 defines measurable performance and quality requirements. Section 5.5 records the hardware and software needed to develop and execute the system. Inspection verifies documents or configuration, demonstration verifies observable user behavior, and test verifies repeatable outcomes against stated criteria.

5.2 CSCI Component Breakdown

The Campus Marketplace Computer Software Configuration Item comprises four Computer Software Components. Each CSC contains Computer Software Units that divide the system into implementable and testable responsibilities.

Component	Contained Units	Responsibility
5.2.1 Client Application CSC	Authentication and Account CSU; Discovery and Search CSU; Vendor Workspace CSU; Booking and Review CSU; Administration Dashboard CSU	Provides the responsive browser interface used by customers, vendors, and administrators.
5.2.2 Marketplace API CSC	Identity and Access CSU; Catalog and Discovery CSU; Availability and Booking CSU; Reviews Moderation and Notifications CSU	Applies marketplace rules and exposes authenticated operations to the client application.
5.2.3 Data and Storage CSC	Account Data CSU; Catalog Data CSU; Request Data CSU; Trust Data CSU; Media Object CSU	Persists structured records and vendor media while maintaining data relationships.
5.2.4 Platform Services CSC	Configuration CSU; Logging and Monitoring CSU; Backup and Recovery CSU	Supports deployment, configuration, operational visibility, and recoverability.

5.3 Functional Requirements by CSC

The following requirements state what the completed system must do. Each entry contains one testable shall statement and a planned verification method. Functional details may be refined through the course change-control process, but requirement identifiers will remain stable whenever possible so later test artifacts can trace back to this specification.

5.3.1 Client Application CSC

The Client Application CSC provides the visible workflows used by customers, vendors, and administrators through a modern browser.

5.3.1.1 Landing Page • Verify: Inspect

The Client Application CSC shall display the purpose of Campus Marketplace and the campus currently in context.

5.3.1.2 Registration Form • Verify: Demo

The Client Application CSC shall display a registration form that collects the information required to create an account.

5.3.1.3 Login Form • Verify: Demo

The Client Application CSC shall display a login form for registered users.

5.3.1.4 Verification Status • Verify: Test

The Client Application CSC shall display the verification state of the signed-in account.

5.3.1.5 Campus Context • Verify: Test

The Client Application CSC shall display the campus associated with the signed-in account.

5.3.1.6 Role Navigation • Verify: Test

The Client Application CSC shall display navigation options appropriate to the active customer, vendor, or administrator role.

5.3.1.7 Keyword Entry • Verify: Demo

The Client Application CSC shall provide a text field for marketplace keyword searches.

5.3.1.8 Category Navigation • Verify: Demo

The Client Application CSC shall provide navigation for browsing marketplace categories.

5.3.1.9 Search Results • Verify: Inspect

The Client Application CSC shall display each search result with the vendor name, category, rating, price indicator, and campus-area location.

5.3.1.10 Search Filters • Verify: Demo

The Client Application CSC shall provide controls for applying the supported marketplace filters.

5.3.1.11 Vendor Profile View • Verify: Demo

The Client Application CSC shall display the selected vendor's profile information and active listings.

5.3.1.12 Listing Detail View • Verify: Demo

The Client Application CSC shall display the selected listing's description, price, images, fulfillment method, and availability.

5.3.1.13 Customer Dashboard • Verify: Demo

The Client Application CSC shall display the customer's upcoming requests, request history, saved vendors, and notifications.

5.3.1.14 Vendor Dashboard • Verify: Demo

The Client Application CSC shall display the vendor's listings, availability, incoming requests, and business performance summary.

5.3.1.15 Availability Calendar • Verify: Demo

The Client Application CSC shall display available service times in a calendar or equivalent time-selection control.

5.3.1.16 Request Form • Verify: Demo

The Client Application CSC shall display a form for submitting an appointment or product reservation request.

5.3.1.17 Request Status • Verify: Test

The Client Application CSC shall display the current status of each appointment or reservation request.

5.3.1.18 Review Form • Verify: Test

The Client Application CSC shall display a review form for an eligible completed transaction.

5.3.1.19 Saved Vendors • Verify: Test

The Client Application CSC shall display the vendors saved by the signed-in customer.

5.3.1.20 System Feedback • Verify: Test

The Client Application CSC shall display a clear confirmation or error message after each submitted form.

5.3.2 Marketplace API CSC

The Marketplace API CSC validates requests, enforces role and ownership rules, coordinates marketplace workflows, and supplies data to the client application.

5.3.2.1 Identity and Access CSU

5.3.2.1.1 Academic Email Registration • Verify: Test

The Identity and Access CSU shall accept account registrations only from email addresses ending in .edu.

5.3.2.1.2 Nonacademic Email Rejection • Verify: Test

The Identity and Access CSU shall reject a registration submitted with an email address that does not end in .edu.

5.3.2.1.3 Verification Message • Verify: Test

The Identity and Access CSU shall send a single-use verification link to each accepted registration email address.

5.3.2.1.4 Verification Gate • Verify: Test

The Identity and Access CSU shall prevent an unverified account from using marketplace functions that require authentication.

5.3.2.1.5 Campus Mapping • Verify: Test

The Identity and Access CSU shall associate a verified account with the campus mapped to its email domain.

5.3.2.1.6 Unique Account • Verify: Test

The Identity and Access CSU shall prevent creation of more than one active account for the same email address.

5.3.2.1.7 Valid Authentication • Verify: Test

The Identity and Access CSU shall create an authenticated session after valid credentials are supplied.

5.3.2.1.8 Invalid Authentication • Verify: Test

The Identity and Access CSU shall deny access after invalid credentials are supplied.

5.3.2.1.9 Session Expiration • Verify: Test

The Identity and Access CSU shall invalidate a session after the configured inactivity period.

5.3.2.1.10 Logout • Verify: Test

The Identity and Access CSU shall invalidate the active session when the user selects logout.

5.3.2.1.11 Password Reset • *Verify: Test*

The Identity and Access CSU shall permit a registered user to reset a password through a time-limited email link.

5.3.2.1.12 Vendor Activation • *Verify: Demo*

The Identity and Access CSU shall permit a verified customer account to activate a vendor profile without creating a second account.

5.3.2.2 Catalog and Discovery CSU

5.3.2.2.1 Vendor Profile Creation • *Verify: Test*

The Catalog and Discovery CSU shall permit a verified user to create one vendor profile.

5.3.2.2.2 Business Identity • *Verify: Inspect*

The Catalog and Discovery CSU shall store a business name and business description for each vendor profile.

5.3.2.2.3 Service Area • *Verify: Test*

The Catalog and Discovery CSU shall store the campus or nearby service area selected by the vendor.

5.3.2.2.4 Listing Creation • *Verify: Demo*

The Catalog and Discovery CSU shall permit a vendor to create a marketplace listing.

5.3.2.2.5 Listing Type • *Verify: Test*

The Catalog and Discovery CSU shall classify each listing as a service or product.

5.3.2.2.6 Listing Category • *Verify: Test*

The Catalog and Discovery CSU shall associate each listing with one supported marketplace category.

5.3.2.2.7 Listing Description • *Verify: Inspect*

The Catalog and Discovery CSU shall store a vendor-provided description for each listing.

5.3.2.2.8 Listing Price • *Verify: Test*

The Catalog and Discovery CSU shall store a nonnegative price or starting price for each listing.

5.3.2.2.9 Listing Images • *Verify: Test*

The Catalog and Discovery CSU shall permit a vendor to attach up to ten images to a listing.

5.3.2.2.10 Fulfillment Location • *Verify: Test*

The Catalog and Discovery CSU shall store the on-campus, nearby, or vendor-specified fulfillment method for each listing.

5.3.2.2.11 Student Discount • *Verify: Test*

The Catalog and Discovery CSU shall permit a vendor to publish an optional discount with an expiration date.

5.3.2.2.12 Listing Visibility • *Verify: Test*

The Catalog and Discovery CSU shall permit a vendor to activate or deactivate a listing.

5.3.2.2.13 Listing Update • *Verify: Test*

The Catalog and Discovery CSU shall permit a vendor to edit a listing owned by that vendor.

5.3.2.2.14 Campus Browse • *Verify: Test*

The Catalog and Discovery CSU shall return active listings associated with the customer's campus context.

5.3.2.2.15 Keyword Search • *Verify: Test*

The Catalog and Discovery CSU shall return active listings whose searchable fields match the submitted keyword.

5.3.2.2.16 Category Filter • *Verify: Test*

The Catalog and Discovery CSU shall restrict search results to the selected category when a category filter is applied.

5.3.2.2.17 Price Filter • *Verify: Test*

The Catalog and Discovery CSU shall restrict search results to the selected price range when a price filter is applied.

5.3.2.2.18 Availability Filter • *Verify: Test*

The Catalog and Discovery CSU shall restrict service results to listings with availability in the selected period.

5.3.2.2.19 Rating Order • *Verify: Test*

The Catalog and Discovery CSU shall order search results by average rating when the rating sort is selected.

5.3.2.2.20 Combined Filters • *Verify: Test*

The Catalog and Discovery CSU shall apply all submitted search filters to the same result set.

5.3.2.2.21 Vendor Portfolio • *Verify: Test*

The Catalog and Discovery CSU shall return all active listings and portfolio images belonging to the selected vendor.

5.3.2.2.22 Saved Vendor Toggle • *Verify: Test*

The Catalog and Discovery CSU shall permit a customer to change the saved status of a vendor profile.

5.3.2.3 Availability and Booking CSU

5.3.2.3.1 Weekly Availability • *Verify: Test*

The Availability and Booking CSU shall permit a vendor to define recurring weekly service availability.

5.3.2.3.2 Availability Exception • *Verify: Test*

The Availability and Booking CSU shall permit a vendor to mark a specific date or time as unavailable.

5.3.2.3.3 Service Duration • *Verify: Test*

The Availability and Booking CSU shall store the appointment duration configured for each service listing.

5.3.2.3.4 Product Quantity • *Verify: Test*

The Availability and Booking CSU shall store the quantity available for each product listing.

5.3.2.3.5 Appointment Request • *Verify: Demo*

The Availability and Booking CSU shall permit a customer to request an appointment for an active service listing.

5.3.2.3.6 Product Reservation • *Verify: Demo*

The Availability and Booking CSU shall permit a customer to request a reservation for an available product listing.

5.3.2.3.7 Appointment Selection • *Verify: Test*

The Availability and Booking CSU shall require a requested service appointment to reference one available date and time.

5.3.2.3.8 Past Time Rejection • *Verify: Test*

The Availability and Booking CSU shall reject an appointment request whose start time is in the past.

5.3.2.3.9 Unavailable Slot Rejection • *Verify: Test*

The Availability and Booking CSU shall reject an appointment request for a time outside the vendor's availability.

5.3.2.3.10 Pending State • *Verify: Test*

The Availability and Booking CSU shall assign pending status to each newly submitted request.

5.3.2.3.11 Request Acceptance • *Verify: Test*

The Availability and Booking CSU shall permit the owning vendor to accept a pending request.

5.3.2.3.12 Request Decline • *Verify: Test*

The Availability and Booking CSU shall permit the owning vendor to decline a pending request.

5.3.2.3.13 Double Booking Prevention • *Verify: Test*

The Availability and Booking CSU shall prevent two accepted service appointments from occupying the same vendor time slot.

5.3.2.3.14 Customer Cancellation • *Verify: Test*

The Availability and Booking CSU shall permit a customer to cancel an eligible request created by that customer.

5.3.2.3.15 Vendor Cancellation • *Verify: Test*

The Availability and Booking CSU shall permit a vendor to cancel an eligible request assigned to that vendor.

5.3.2.3.16 Reschedule Request • *Verify: Test*

The Availability and Booking CSU shall permit a customer to propose a new available time for an existing service request.

5.3.2.3.17 Request History • *Verify: Test*

The Availability and Booking CSU shall return upcoming and past requests for the authenticated user.

5.3.2.3.18 Status History • *Verify: Inspect*

The Availability and Booking CSU shall record each status transition for an appointment or reservation request.

5.3.2.4 Reviews Moderation and Notifications CSU

5.3.2.4.1 Review Eligibility • *Verify: Test*

The Reviews Moderation and Notifications CSU shall permit a review only after the related request reaches completed status.

5.3.2.4.2 Rating Scale • *Verify: Test*

The Reviews Moderation and Notifications CSU shall require each review to include an integer rating from one through five.

5.3.2.4.3 Review Comment • *Verify: Test*

The Reviews Moderation and Notifications CSU shall permit each review to include an optional text comment.

5.3.2.4.4 Single Review • *Verify: Test*

The Reviews Moderation and Notifications CSU shall prevent more than one active customer review for the same completed request.

5.3.2.4.5 Average Rating • *Verify: Test*

The Reviews Moderation and Notifications CSU shall calculate a vendor's displayed rating from active customer ratings.

5.3.2.4.6 Vendor Response • *Verify: Test*

The Reviews Moderation and Notifications CSU shall permit a vendor to post one response to a review on that vendor's profile.

5.3.2.4.7 Content Report • *Verify: Demo*

The Reviews Moderation and Notifications CSU shall permit a verified user to report a vendor profile, listing, or review.

5.3.2.4.8 Moderation Queue • *Verify: Test*

The Reviews Moderation and Notifications CSU shall place each submitted report in an administrator moderation queue.

5.3.2.4.9 Content Moderation • *Verify: Test*

The Reviews Moderation and Notifications CSU shall permit an administrator to hide or restore reported content.

5.3.2.4.10 Account Moderation • *Verify: Test*

The Reviews Moderation and Notifications CSU shall permit an administrator to deactivate or reactivate an account.

5.3.2.4.11 Moderation Audit • *Verify: Inspect*

The Reviews Moderation and Notifications CSU shall record the administrator, timestamp, target, and result of each moderation action.

5.3.2.4.12 New Request Notice • *Verify: Test*

The Reviews Moderation and Notifications CSU shall notify a vendor when a new request is submitted.

5.3.2.4.13 Status Change Notice • *Verify: Test*

The Reviews Moderation and Notifications CSU shall notify affected users when a request status changes.

5.3.2.4.14 Appointment Reminder • *Verify: Test*

The Reviews Moderation and Notifications CSU shall send a reminder before an accepted service appointment.

5.3.2.4.15 Notification Preference • *Verify: Test*

The Reviews Moderation and Notifications CSU shall permit a user to enable or disable optional email notifications.

5.3.3 Data and Storage CSC

The Data and Storage CSC preserves marketplace records, enforces stored-data relationships, and supports authorized data lifecycle operations.

5.3.3.1 Account Persistence • *Verify: Inspect*

The Data and Storage CSC shall persist user identity, campus association, role, and profile data.

5.3.3.2 Catalog Persistence • *Verify: Inspect*

The Data and Storage CSC shall persist vendor, listing, category, price, discount, and media metadata.

5.3.3.3 Request Persistence • *Verify: Inspect*

The Data and Storage CSC shall persist availability, appointment, reservation, and status-history data.

5.3.3.4 Trust Persistence • *Verify: Inspect*

The Data and Storage CSC shall persist reviews, reports, notification preferences, and moderation records.

5.3.3.5 Referential Integrity • *Verify: Test*

The Data and Storage CSC shall reject a stored relationship that references a nonexistent parent record.

5.3.3.6 Atomic Booking • *Verify: Test*

The Data and Storage CSC shall commit an accepted booking and its reserved time slot as one transaction.

5.3.3.7 User Data Export • *Verify: Test*

The Data and Storage CSC shall produce a machine-readable export of the authenticated user's stored account data.

5.3.3.8 Account Deletion • *Verify: Test*

The Data and Storage CSC shall remove or anonymize personal account data after an authorized deletion request.

5.3.4 Platform Services CSC

The Platform Services CSC provides the operational functions required to configure, observe, and recover the deployed system.

5.3.4.1 Health Check • *Verify: Test*

The Platform Services CSC shall expose a health-check result for the deployed API.

5.3.4.2 Structured Logging • *Verify: Inspect*

The Platform Services CSC shall record structured logs for authentication, API, and background-task failures.

5.3.4.3 External Configuration • *Verify: Inspect*

The Platform Services CSC shall load environment-specific configuration outside the committed source code.

5.3.4.4 Safe Error Response • *Verify: Test*

The Platform Services CSC shall return user-safe error messages without exposing stack traces or secrets.

5.3.4.5 Scheduled Backup • *Verify: Inspect*

The Platform Services CSC shall initiate a database backup on the configured schedule.

5.4 Performance Requirements by CSC

The performance requirements define measurable operating characteristics for the completed system. The reference load is the controlled course test environment with representative data and up to 100 simultaneous authenticated users.

5.4.1 Client Application CSC

5.4.1.1 Initial Load • *Verify: Test*

The Client Application CSC shall become interactive within three seconds for at least 95 percent of measured loads over a 10 Mbps connection.

5.4.1.2 Interaction Feedback • *Verify: Test*

The Client Application CSC shall acknowledge a local button or form interaction within 200 milliseconds.

5.4.1.3 Responsive Layout • *Verify: Test*

The Client Application CSC shall remain usable at viewport widths from 320 through 1440 pixels.

5.4.1.4 Accessibility • *Verify: Test*

The Client Application CSC shall conform to WCAG 2.1 Level AA for the user flows included in the release demonstration.

5.4.1.5 Browser Support • *Verify: Test*

The Client Application CSC shall complete the critical user flows in the two most recent major versions of Chrome, Safari, Firefox, and Edge.

5.4.1.6 Horizontal Fit • *Verify: Test*

The Client Application CSC shall avoid horizontal page scrolling at each supported viewport width.

5.4.2 Marketplace API CSC

5.4.2.1 Read Response Time • *Verify: Test*

The Marketplace API CSC shall complete at least 95 percent of standard read requests within two seconds under the reference load.

5.4.2.2 Write Response Time • *Verify: Test*

The Marketplace API CSC shall complete at least 95 percent of standard write requests within three seconds under the reference load.

5.4.2.3 Search Response Time • *Verify: Test*

The Marketplace API CSC shall return the first page of search results within three seconds under the reference load.

5.4.2.4 Concurrent Users • *Verify: Test*

The Marketplace API CSC shall support 100 simultaneous authenticated users without violating the response-time requirements.

5.4.2.5 Verification Delivery • *Verify: Test*

The Marketplace API CSC shall submit a verification email to the notification provider within ten seconds of an accepted registration.

5.4.2.6 Input Validation • *Verify: Test*

The Marketplace API CSC shall reject every malformed request in the defined API validation suite.

5.4.2.7 Authentication Throttling • *Verify: Test*

The Marketplace API CSC shall limit repeated failed login attempts according to the configured rate-limit policy.

5.4.3 Data and Storage CSC

5.4.3.1 Backup Frequency • *Verify: Inspect*

The Data and Storage CSC shall create at least one recoverable database backup every 24 hours while the system is deployed.

5.4.3.2 Recovery Point • *Verify: Test*

The Data and Storage CSC shall limit recoverable data loss to no more than 24 hours after a declared database failure.

5.4.3.3 Recovery Time • *Verify: Test*

The Data and Storage CSC shall restore the database to a usable state within four hours of a declared database failure.

5.4.3.4 Concurrent Booking Integrity • *Verify: Test*

The Data and Storage CSC shall create no duplicate accepted appointment when twenty concurrent requests target the same time slot.

5.4.3.5 Common Query Time • *Verify: Test*

The Data and Storage CSC shall complete at least 95 percent of indexed queries in the performance test suite within 500 milliseconds.

5.4.4 Platform Services CSC

5.4.4.1 Demonstration Availability • *Verify: Test*

The Platform Services CSC shall provide 99 percent availability during the scheduled final demonstration window.

5.4.4.2 Transport Security • *Verify: Inspect*

The Platform Services CSC shall accept production client traffic only through HTTPS using TLS 1.2 or later.

5.4.4.3 Secret Storage • *Verify: Inspect*

The Platform Services CSC shall store production credentials in a managed secret store rather than in source-controlled files.

5.4.4.4 Least Privilege • *Verify: Inspect*

The Platform Services CSC shall grant each deployed service only the permissions required by its documented functions.

5.4.4.5 Failure Visibility • *Verify: Test*

The Platform Services CSC shall make a logged application failure visible to the project team within one minute.

5.4.4.6 Release Vulnerabilities • *Verify: Inspect*

The Platform Services CSC shall have no unresolved critical-severity dependency vulnerability at code freeze.

5.5 Project Environment Requirements

5.5.1 Development Environment Requirements

The development environment must support local frontend and backend execution, automated testing, database work, source control, and access to the selected cloud services. Equivalent operating systems and editors may be used when they support the same required tools.

Category	Requirement	Purpose
Computer	64-bit laptop with at least 8 GB RAM, four logical CPU cores, and 20 GB free	Local development, builds, and automated tests
Network	Reliable broadband internet connection	Repository, cloud, package, and email-service access
Operating System	Current macOS, Windows, or Linux release	Supported local development environment
Source Control	Git and a shared GitHub repository	Version control, review, issue tracking, and CI
Frontend	Node.js, npm, React, and TypeScript	Responsive browser client

Additional Development Tools

Category	Requirement	Purpose
Backend	Python, FastAPI, and an ASGI development server	REST API and business rules
Database	PostgreSQL and a database administration client	Schema development and local data testing
Testing	Pytest, Vitest, React Testing Library, and Postman or equivalent	Unit, component, API, and integration testing
Editor	Visual Studio Code, IntelliJ, or equivalent IDE	Source editing, debugging, and review
Design	Figma, Canva, or equivalent diagram and wireframe tool	User flows, interface designs, and documentation figures
Cloud Access	Authorized AWS development account	Hosted integration, deployment, storage, and monitoring

5.5.2 Execution Environment Requirements

The execution environment includes the user's browser and the cloud-hosted application services used during demonstrations and acceptance testing.

Category	Requirement	Purpose
Client Device	Internet-connected computer, tablet, or smartphone with a viewport at least 320 pixels wide	Access the responsive web application
Browser	One of the supported Chrome, Safari, Firefox, or Edge versions	Render and operate the client application
Frontend Hosting	AWS Amplify or an equivalent managed static-web hosting service	Serve the React application through HTTPS
API Hosting	AWS App Runner or an equivalent managed Python web-service platform	Execute the FastAPI application
Authentication	Managed identity service capable of email verification	Verify .edu addresses and manage authenticated sessions

Data and Operations Services

Category	Requirement	Purpose
Database	Managed PostgreSQL instance	Persist marketplace and audit data
Object Storage	Managed object storage with private upload controls	Store vendor and listing images
Email	Transactional email provider	Deliver verification, status, and reminder messages
Monitoring	Centralized application logs and health monitoring	Support failure detection and diagnosis

No unusual end-user hardware is required. The development team must obtain access to the selected managed services before integration testing begins. If a named cloud service becomes unavailable or exceeds the project budget, the team may substitute a service with equivalent authentication, hosting, database, storage, email, and monitoring capabilities without changing the system requirements.